

Chapter 8.2 — Top 10 Exam-Ready Questions & Answers

Q1. What are the major differences between Formal Design Reviews (FDRs) and Peer Reviews in terms of participants and authority?

বাংলা অর্থ

FDR এবং Peer Review-এর মূল পার্থক্য হলো কারা অংশ নেয় এবং তাদের authority কতটুকু।

FDR:

- Senior people / superior-level people অংশ নেয়।
- Customer representative থাকতে পারে।
- তারা design **approve করতে পারে।**
- Approval দিলে project next phase-এ যেতে পারে।

Peer Review:

- একই level-এর developer/colleague অংশ নেয়।
- তাদের মূল কাজ **error খুঁজে বের করা।**
- Design approve করার authority নেই।

Exam-Ready Answer

The major difference between FDRs and peer reviews lies in their participants and authority.

In FDRs, participants are generally senior to the project team and may include customer representatives. The FDR team has the authority to approve the design document so that the project can proceed.

In peer reviews, participants are generally peers from the same or other departments. They focus mainly on detecting errors and deviations from standards and do not have formal approval authority.

□ **মনে রাখো**

FDR = Senior + Approval

Peer Review = Equal-level + Error Detection

Q2. How do Inspections and Walkthroughs differ in terms of formality and their primary objectives?

বাংলা অর্থ

Inspection এবং Walkthrough দুটোই Peer Review-এর type, কিন্তু:

Inspection → বেশি Formal

Walkthrough → কম Formal

Inspection-এর main focus হলো **error correction/action** এবং SQA improve করা।

Walkthrough-এর main focus হলো document সম্পর্কে **comments/feedback দেওয়া**।

Exam-Ready Answer

Both inspections and walkthroughs are peer review methods, but they differ in formality and objectives.

Inspections are more formal and emphasize defect detection, corrective action, and improvement of software development methods.

Walkthroughs are less formal and mainly focus on providing comments and feedback on the reviewed document.

Therefore, inspections generally contribute more strongly to the overall SQA process.

□ **Shortcut**

Inspection = Formal + Corrective Action

Walkthrough = Informal + Comments

Q3. Describe the differing roles of the "Author" and the "Presenter" in Inspections versus Walkthroughs.

বাংলা অর্থ

এখানে মূল পার্থক্য হলো কে document/code present করবে।

Inspection:

Author এবং Presenter সাধারণত আলাদা ব্যক্তি।

কারণ Author নিজে present করলে bias হতে পারে।

Walkthrough:

Author নিজেই Presenter।

কারণ author document সম্পর্কে সবচেয়ে ভালো জানে।

Exam-Ready Answer

In an inspection, the presenter is selected by the moderator and normally should not be the author of the document. This helps maintain objectivity.

In a walkthrough, the author normally acts as the presenter because the author is most familiar with the document and its design logic.

□ মনে রাখো

Inspection → Author ≠ Presenter

Walkthrough → Author = Presenter

Q4. Who are the "Specialized Professionals" typically involved in an Inspection compared to a Walkthrough?

বাংলা অর্থ

Inspection-এ মূলত **technical development people** বেশি গুরুত্বপূর্ণ।

যেমন:

- Designer
- Coder
- Tester

Walkthrough-এ:

- Standards expert
- Maintenance expert
- User representative

Exam-Ready Answer

In inspections, specialized professionals usually include a designer or systems analyst, a coder or implementer, and a tester, preferably the leaders of the respective teams.

In walkthroughs, specialized participants may include a standards enforcer, maintenance experts, and a user representative.

সহজে মনে রাখো

Inspection → **Design + Code + Test**

Walkthrough → **Standard + Maintenance + User**

Q5. Contrast the preparation requirements for a review team prior to an Inspection versus a Walkthrough.

বাংলা অর্থ

Inspection-এর আগে বেশি preparation করতে হয়।

Inspection:

Team members:

- Document ভালোভাবে পড়বে
- Comments আগে থেকে তৈরি করবে
- Checklist ব্যবহার করবে

Walkthrough:

- শুধু document-এর general idea নিলেই হয়
- আগে থেকে detailed comments তৈরি করা সাধারণত বাধ্যতামূলক নয়

Exam-Ready Answer

Inspection preparation is more thorough than walkthrough preparation.

Before an inspection, participants must carefully read the document, identify and list comments, and often use a checklist specific to the document type.

Before a walkthrough, participants usually perform only a brief reading to obtain a general understanding, and advance preparation of detailed comments is generally not required.

□ Shortcut

Inspection → Prepare deeply

Walkthrough → Prepare briefly

Q6. What is the fundamental restriction placed on discussions during a Peer Review Session?

বাংলা অর্থ

Peer Review meeting-এ মূল কাজ হলো **error identify করা**।

Meeting-এর মধ্যে:

✗ Error কীভাবে fix করব—এটা নিয়ে discussion করা যাবে না।

✓ Error কোথায় এবং কী ধরনের—সেটা identify করা যাবে।

Exam-Ready Answer

The fundamental restriction is that discussions during a peer review should be limited to the **identification of errors**.

Participants may comment on the document or other comments, but they should not discuss possible solutions during the review session. Solutions should be handled separately after the review.

□ মনে রাখো

Review Meeting = Find Error

Not = Solve Error

Q7. What is the most fundamental differentiating element between Inspections and Walkthroughs in the Post-Review Session?

বাংলা অর্থ

Inspection এবং Walkthrough-এর সবচেয়ে বড় difference **review-এর পরে দেখা যায়**।

Inspection:

Review শেষ হওয়ার পর:

- Error fix করতে হবে
- Correction verify করতে হবে
- Follow-up হবে
- Report controlling authority-কে দেওয়া হতে পারে

Walkthrough:

এই ধরনের **strict post-review follow-up** সাধারণত থাকে না।

Exam-Ready Answer

The most fundamental difference appears in the post-review activities.

An inspection does not end with the review session. Strict follow-up activities are performed to ensure that all identified defects are promptly and effectively corrected.

Inspection reports may also be sent to a controlling authority for further analysis.

□ মনে রাখো

Inspection → Strict Follow-up

Walkthrough → Less Formal Follow-up

Q8. What did the Cusumano study at Fujitsu indicate regarding the effectiveness of code inspections?

বাংলা অর্থ

এটা খুব গুরুত্বপূর্ণ একটা study।

Fujitsu-তে 1977–1982 সালের study দেখিয়েছে যে Design Review এবং Code Inspection বাড়ালে software defect অনেক কমে যায়।

তাদের data অনুযায়ী:

- Code inspection + design review → **15% থেকে 30%**
- Software testing → **85% থেকে 30%**

- Defects per 1000 lines → **0.19 থেকে 0.02**

অর্থাৎ inspection/review-এর মাধ্যমে **অনেক আগে থেকেই defect ধরা সম্ভব**, ফলে final software-এর defect dramatically কমে।

Exam-Ready Answer

The Cusumano study at Fujitsu from 1977 to 1982 demonstrated the effectiveness of design reviews and code inspections.

As the share of design reviews and code inspections increased from 15% to 30%, while software testing decreased from 85% to 30%, the number of defects per 1000 lines of maintained code dropped significantly from **0.19 to 0.02**.

This indicates that early reviews and inspections can substantially improve software quality.

□ মূল শিক্ষা

More Inspection/Review → Fewer Defects

Q9. Which sections of a software project are recommended to be included in peer reviews, and which should be omitted?

বাংলা অর্থ

সব code/document peer review করার দরকার নেই।

Review করা উচিত:

- Complex logic
- Critical section
- New environment-এর code
- New/inexperienced developer-এর code

Review না করলেও চলে:

- Simple code
- আগের project-এ already reviewed similar code
- Fault হলেও functionality-তে significant impact করবে না এমন section
- Reused/repeated code

Exam-Ready Answer

Sections recommended for peer review include:

- Complicated or complex logic.
- Critical sections where defects could severely affect system capability.
- Sections dealing with new environments.
- Sections designed or developed by new or inexperienced team members.

Sections that may be omitted include:

- Straightforward sections without significant complexity.
- Sections already reviewed in similar previous projects.
- Sections unlikely to significantly affect functionality if defective.
- Reused or repeated design and code.

□ **সহজে**

Complex + Critical + New + Inexperienced → Review

Simple + Reused + Already Reviewed → May Skip

Q10. Under what specific situations do experts suggest the participation of an external expert in reviews?

বাংলা অর্থ

কখন বাইরের একজন expert review-এ আনা উচিত?

যখন:

1. Team-এর ভিতরে required specialist নেই।
2. Temporarily required expert available নেই।
3. Senior professionals-এর মধ্যে বড় disagreement হয়েছে।
4. Organization ছোট এবং review team বানানোর মতো যথেষ্ট লোক নেই।

Exam-Ready Answer

An external expert may participate in a review under the following situations:

1. When the organization lacks sufficient professional expertise in a specialized area.
2. When there is a temporary shortage of qualified internal professionals.
3. When major disagreements among senior professionals prevent a decision.
4. When a small organization does not have enough suitable personnel to form an effective review team.

□ Shortcut

No Expertise → No Availability → Disagreement → Small Team

☆ Bonus Short Notes

1. Peer Review Detection Efficiency

বাংলা অর্থ

Peer Review কতটা effective সেটা measure করার একটি উপায় হলো:

একটি defect খুঁজে বের করতে গড়ে কত ঘণ্টা সময় লাগছে।

Formula

Detection Efficiency = Total Review Hours ÷ Number of Defects Detected

Exam-Ready Answer

Peer Review Detection Efficiency measures the average number of hours required to detect one defect during a peer review. It is calculated by dividing total review hours by the number of defects detected.

2. Inspection Checklist & Defect Frequency Tables

বাংলা অর্থ

Inspection-এর আগে:

Checklist → কী কী common error খুঁজব

Defect Frequency Table → অতীতে কোন ধরনের error বেশি হয়েছে

এগুলো reviewer-কে সম্ভাব্য defect concentration area খুঁজে পেতে সাহায্য করে।

Exam-Ready Answer

Inspection checklists and defect frequency tables are important parts of inspection infrastructure.

Checklists identify common defects to look for in specific document or programming languages, while defect frequency tables show the types of defects that occurred most frequently in previous reviews.

□ সহজে

Checklist = কী খুঁজব

Frequency Table = কোন error বেশি হয়

3. Walkthrough Scribe vs Inspection Scribe

বাংলা অর্থ

Scribe হলো যে review meeting-এর important information/error লিখে রাখে।

Walkthrough Scribe

Record করে:

- Error location
- Error description
- Error type

Inspection Scribe

উপরের সবকিছুর পাশাপাশি:

- Defect severity-ও record করে।

এটা পরে statistical analysis-এর কাজে লাগে।

Exam-Ready Answer

In a walkthrough, the scribe records each identified error, including its location, description, and type.

In an inspection, the scribe records the same information and additionally records the estimated **severity** of each defect for later statistical analysis.

Chapter 8.2 — এক নজরে Revision

Topic	সহজ অর্থ
FDR	Senior participants + Approval authority
Peer Review	Equal-level participants + Error detection
Inspection	Formal + Corrective action
Walkthrough	Less formal + Comments
Inspection Presenter	Usually not Author
Walkthrough Presenter	Author
Inspection Team	Designer + Coder + Tester
Walkthrough Team	Standards + Maintenance + User
Inspection Preparation	Detailed
Walkthrough Preparation	Brief
Peer Review Discussion	Error identification only
Inspection Post-review	Strict follow-up
Fujitsu Study	More inspection → fewer defects
Review Priority	Complex + Critical + New
External Expert	Lack of expertise/availability, disagreement, small team
Detection Efficiency	Review hours per defect
Scribe	Records review defects

★ সবচেয়ে গুরুত্বপূর্ণ Difference গুলো

FDR vs Peer Review

FDR → Approval আছে

Peer Review → Approval নেই

Inspection vs Walkthrough

Inspection → Formal + Author usually doesn't present + Strict follow-up

Walkthrough → Less formal + Author presents + Mainly comments

Inspection Discussion

Find defects, don't solve them during the session.

Fujitsu Study

Inspection/Review $\uparrow \rightarrow$ Defects $\downarrow$